

Conviction Engine Fixes, Missing Implementations, & Feature Additions

To: Divyanshu

Asset Live Test Case: PYPL (PayPal Holdings, Inc.)

Target Verification Assets: KXS.TO (Canadian data source path testing) or MSFT (Large-cap US yfinance path testing)

Please systematically execute the fixes, missing implementations, new features, and frontend updates outlined below. Once the **P1** items are completed, please run the verification engine on PYPL to confirm the conviction gap moves. Additionally, perform a comprehensive manual audit on KXS.TO or MSFT, ensuring every BQ dimension, valuation tax component, OEY, and PE percentile is pulled and computed correctly before marking the project production-ready.

— SIZING AND fd_direction — OPERATIONAL LOGIC —

The fd_direction (Fundamental Direction) flag does **NOT** alter the final conviction verdict tier itself. Instead, it systematically modulates the **SIZE PERCENTAGE** allocated within that specific tier:

Verdict Tier	Base Size	fd = POSITIVE	fd = NEGATIVE
MAX CONVICTION	100%	100% (<i>no change</i>)	100% (<i>no change</i>)
TACTICAL BUY	75%	→ 85%	→ 60%
REDUCED BUY	40%	→ 50%	→ 25%
CANCEL BUY	0%	0%	0%

- **PYPL Target Case:** Corrected conviction +8 (**MAX CONVICTION**) + fd=POSITIVE $\rightarrow$ **Size = 100%**.
- **PYPL Current Buggy State:** Conviction +1 (**CANCEL BUY**) $\rightarrow$ **Size = 0%** regardless of fundamental direction status.

SECTION 1 — DATA & FORMULA BUGS (P1 — Fix Immediately)

BUG 1: trailingPE Returning 0 or Null for PYPL

- **Symptom:** Displays as "Trailing PE 0x" in the Financial Strength (FS) tab. The PE 20-Year percentile calculates as the 0th percentile.
- **Consequence:** The system applies an erroneous -2 PE percentile penalty instead of 0, costing the model 2 conviction points. (Real PYPL PE is ~10.61x and Real EPS TTM is ~\$5.33).
- **Fix:** Implement a manual EPS/PE calculation cascade fallback:

Python

```
eps_ttm = info.get('trailingEps')
```

```
if not eps_ttm or eps_ttm <= 0:
```

```
    # Fallback: Sum last 4 quarters of Net Income from the income statement
```

```
    net_income_ttm = income_stmt.loc['Net Income'].iloc[:4].sum()
```

```
    shares = info.get('sharesOutstanding') or info.get('impliedSharesOutstanding')
```

```
    eps_ttm = net_income_ttm / shares if shares else None
```

```
pe_ttm = (current_price / eps_ttm) if eps_ttm and eps_ttm > 0 else None
```

- **CRITICAL EDGE CONDITION RULE:** If pe_ttm evaluates to None or 0 after all programmatic fallbacks, **default the valuation tax PE component to 0 (neutral). NEVER** apply a worst-case penalty (-2) for missing data. This applies globally to all data fields across the engine: *Data Failure $\rightarrow$ 0 penalty (Neutral)*.

BUG 2: Owner Earnings Yield (OEY) Returning 0% for PYPL

- **Symptom:** Displays as "0%" in the Financial Strength tab. (Real TTM Free Cash Flow is ~\$5.503B, yielding an actual OEY of 11.0% against a ~\$49.89B market cap).
- **Consequence:** OEY displays as 0% instead of matching/exceeding the 4% compounder threshold, blocking an evaluation of "Strong".
- **Fix:** Implement a three-tier Free Cash Flow cascade fallback:

Python

```

fcf = info.get('freeCashflow')                    # Try 1: Annual field
if not fcf or fcf <= 0:
    cf = stock.cashflow
    if 'Free Cash Flow' in cf.index:
        fcf = cf.loc['Free Cash Flow'].iloc[:4].sum()    # Try 2: TTM Sum
if not fcf or fcf <= 0:
    # Try 3: Manual Calculation (Operating Cash Flow minus Capex)
    ocf = cf.loc['Operating Cash Flow'].iloc[:4].sum()
    capex = abs(cf.loc['Capital Expenditure'].iloc[:4].sum())
    fcf = ocf - capex

```

oe = fcf / market_cap if market_cap > 0 else 0

BUG 3: Divergence Signal Returning 0 (Target State: +2)

- **Symptom:** Signal fires as 0 despite all criteria being met.
- **Conditions Met:**
 1. Current Price (\$56.56) is 28.9% below the 52-Week High (\$79.50) [$\geq 15\%$ Criteria $\checkmark$]
 2. Asset has been $\geq 15\%$ below its 52-Week high for 5+ months (since late 2025/Feb 12 2026 low) [≥ 60 days Minimum $\checkmark$]
 3. fd_direction $\neq$ negative [$\checkmark$]
- **Root Cause:** The days_below counter is failing to persist across daily scheduled runs or is resetting during execution recalculations.
- **Fix:**
 1. Add active runtime logging:

Python

```

print(f'[divergence] {ticker}: price={current_price}, 52Wh={high}, pct_below={pct:.1%},
days={days}')

```

2. Persist days_below_high within the daily conviction_store JSON file to maintain continuity across runs.

SECTION 2 — MISSING DIMENSION IMPLEMENTATIONS (P2)

— DIM 8: CEO QUALITY (-1 → +2) —

- **Core Logic:** Fully objective Total Shareholder Return (TSR) calculation for established CEOs, supplemented by agent web search for newer tenures.
- **Step 1 — Objective TSR Calculation (CEOs > 12 Months):** Pull ceo_start_date from SEC DEF14A proxies (stored in conviction_store per ticker).

Python

```
tsr_vs_spy = compute_tsr(ticker, ceo_start_date) - compute_tsr('SPY', ceo_start_date)
```

```
if ceo_tenure_months < 12:
```

```
    tsr_score = 5 # Neutral base, mechanical transition penalty applied in Step 4
```

```
elif tsr_vs_spy > 0.30:
```

```
    tsr_score = 9 # Outperformed S&P 500 by 30%+
```

```
elif tsr_vs_spy > 0.10:
```

```
    tsr_score = 7 # Outperformed by 10-30%
```

```
elif tsr_vs_spy > -0.10:
```

```
    tsr_score = 5 # In-line with market
```

```
elif tsr_vs_spy > -0.25:
```

```
    tsr_score = 3 # Underperformed
```

```
else:
```

```
    tsr_score = 1 # Badly underperformed
```

- **Step 2 — Supplementary Agent Web Search:** Look up historical qualitative evidence from *HBR Best-Performing CEOs ranking (hbr.org)*, *Glassdoor CEO approval % (glassdoor.com)*, and LinkedIn domain expertise. Blend the quantitative tsr_score with qualitative search context using anchored confidence scoring.

- **Step 3 — Score Mapping:**

- **8–10 $\rightarrow$ +2** (*Exceptional: Founder, 30+ year domain expert, outperformed S&P by 30%+*)
- **6–7 $\rightarrow$ +1** (*Good: Solid track record, respected, roughly in-line with market*)
- **3–5 $\rightarrow$ 0** (*Adequate: No major failures, but no standout quality*)
- **0–2 $\rightarrow$ -1** (*Poor: Value-destructive decisions, material underperformance*)

- **Step 4 — New CEO Transition Penalty (Mandatory Mechanical Rule):**

Python

if ceo_tenure_months < 12:

score = max(score - 1, -1) # Floor at -1. Eliminates guesswork/agent bias.

- **PYPL Case Profile:** Enrique Lores (Started March 1, 2026 $\rightarrow$ ~4.5 months tenure). Background is hardware (HP Inc.), not fintech. TSR cannot be reliably measured. Base score defaults to neutral (5/10 = 0). Mechanical transition penalty shifts the final dimension score to **-1**.

— **DIM 9: CAPITAL ALLOCATION (-1 $\rightarrow$ +2)** —

- **Primary Source:** Extract historical share data via yfinance to assess float reductions over the trailing 12 months.

Python

buyback_pct = (shares_12m_ago - shares_now) / shares_12m_ago

- **Secondary/Tertiary Sources:** Cross-reference calculations via SEC EDGAR Form 4 filings and 10-Q share repurchase footnotes. Use agent web search for historical M&A capital deployment efficiency.
- **Score Mapping:**
 - **8–10 $\rightarrow$ +2** (*$\geq 1\%$ shares retired TTM via an active buyback program*)
 - **6–7 $\rightarrow$ +1**
 - **3–5 $\rightarrow$ 0**

- **0–2 $\rightarrow$ -1**
- **PYPL Case Profile:** ~100M shares retired in the trailing 12 months (882M vs ~950M prior). Represents a 7.1% float reduction, which comfortably clears the threshold for an exceptional score. Target dimension score: **+2**.

— **DIM 10: COMPETITIVE MOAT (-1 $\rightarrow$ +2)** —

- **Core Formula:** Gartner Rank + Proprietary IP + Switching Costs + Customer Lock-in.
- **Agent Prompt Strategy (Adversarial Framework — Threats Evaluated First):**

Plaintext

System = "You are an adversarial investment analyst. First identify the 3 most significant threats to this company's competitive position, including any new entrants, technology disruption, or margin erosion risks. Only then score the overall moat strength. Apply a conservative bias. Return JSON: {score:<0-10>, rationale:<120chars>, threats:[str,str,str], moat_sources:[str], evidence_against:<str>, sources:[str], confidence:<float>}"

- **Sector-Specific Verification Sources:**
 - *Tech/Software:* Gartner Magic Quadrant (gartner.com) + G2 (g2.com)
 - *Consumer/Fintech:* Market share trends (Total Payment Volume - TPV %), active account growth rates, take rate stability (pricing power), and Nilson Report data.
 - *Industrial:* Market share stability + customer concentration matrices derived from 10-K filings.
 - *Healthcare:* Patent cliff calendars, regulatory approval timelines, and formulary positions.
 - *Income/Utility:* Regulatory exclusivity boundaries and replacement infrastructure replication costs.
- **Score Mapping:** 8–10 $\rightarrow$ +2 | 6–7 $\rightarrow$ +1 | 3–5 $\rightarrow$ 0 | 0–2 $\rightarrow$ -1. *Default to 0 if agent confidence score < 0.7.*
- **PYPL Case Profile:** Adversarial queries capture Morningstar data noting slipping market share and heightened large-cap ecosystem competition. Target dimension score: **0** (*Adequate but eroding*).

— DIM 11: MACRO TAILWIND (-1 → +2) —

- **Core Logic:** Agent web search utilizing anchored confidence scoring.
- **Implementation Rules:** The prompt must follow the adversarial pattern, seeking macro headwinds *first*. Map outputs directly to scores of -1, 0, +1, or +2 without an intermediary 0–10 scale. Default the score to 0 if agent confidence falls below 0.7.
- **Python Engine Check:** If $\text{len}(\text{sources}) < 2$, programmatically cap confidence at 0.35 regardless of what the LLM agent reports, forcing a neutral score of 0. The `evidence_against` output string field is mandatory.
- **PYPL Case Profile:** Digital payment volume represents a structural tailwind, balanced by consumer spending pressures linked to high interest rates and tariff inflation. Target dimension score: **0** or **+1**.

— DIM 15: REINVESTMENT RUNWAY (-1 → +1) —

- **Core Logic:** Agent searches for management-stated Total Addressable Market (TAM).
- **Source Priority Hierarchy:**
 1. Most recent corporate Investor Day presentations.
 2. Earnings call transcripts containing official TAM guidance citations.
 3. Grand View Research / MarketsandMarkets / Gartner forecast reports.
- **Execution Rules:** TAM values must always be appended alongside their matching source inside the output rationale string. If TAM cannot be identified, log a system warning and default the score to 0.
- **Score Mapping:**
 - $\frac{\text{TAM}}{\text{Revenue}} > 10 \times \rightarrow +1$
 - $3 \times \text{to } 10 \times \rightarrow 0$
 - $< 3 \times \rightarrow -1$
- **PYPL Case Profile:** Global digital payments TAM stands in the multi-trillion range against a TTM revenue base of \$33.7B ($> 10 \times$ ratio). Target dimension score: **+1**.

SECTION 3 — THE FIVE FUNDAMENTAL DIRECTION (FD) VOTES

These 5 votes run directly on raw yfinance underlying data structures—**no LLM agent parsing is required.**

Python

Vote 1 — Revenue Accelerating

```
qtrs = quarterly_income_stmt.loc['Total Revenue'].sort_index()
```

```
growth_rates = qtrs.pct_change(quarters=4).dropna() # YoY quarterly performance
```

```
last3 = growth_rates.tail(3).values
```

```
revenue_accelerating = bool(last3[-1] > last3[-2] > last3[-3])
```

```
vote_1 = 'positive' if revenue_accelerating else 'stable'
```

Vote 2 — Margins Expanding ≥ 100 bp YoY

```
gm_now = (revenue_ttm - cogs_ttm) / revenue_ttm
```

```
gm_1y = (revenue_1y - cogs_1y) / revenue_1y
```

```
vote_2 = 'positive' if (gm_now - gm_1y) >= 0.01 else ('negative' if (gm_now - gm_1y) <= -0.01 else 'stable')
```

Vote 3 — FCF Growing YoY

```
vote_3 = 'positive' if (fcf_ttm > fcf_prior_year and fcf_margin_expanding) else ('negative' if fcf_ttm < fcf_prior_year * 0.95 else 'stable')
```

Vote 4 — EPS Revisions $\geq 3\%$ Upward in 90 Days

Compare current consensus EPS estimate for next quarter vs value stored 90 days ago in conviction_store

```
eps_now = stock.earnings_estimate['mean'].iloc[0]
```

```
eps_90d = stored_eps_estimate_from_conviction_store
```

```
revision_pct = (eps_now - eps_90d) / abs(eps_90d) if eps_90d else 0
```

```
vote_4 = 'positive' if revision_pct >= 0.03 else ('negative' if revision_pct <= -0.03 else 'stable')
```

Vote 5 — Buybacks $\geq 1\%$ Shares Retired TTM

shares_change = (shares_12m_ago - shares_now) / shares_12m_ago

vote_5 = 'positive' if shares_change ≥ 0.01 else ('negative' if shares_change < 0 else 'stable')

Final Majority Rule Consensus Consolidation

fd_positive_count = sum(1 for v in [vote_1, vote_2, vote_3, vote_4, vote_5] if v == 'positive')

fd_negative_count = sum(1 for v in [vote_1, vote_2, vote_3, vote_4, vote_5] if v == 'negative')

fd_direction = 'positive' if fd_positive_count ≥ 3 else ('negative' if fd_negative_count ≥ 3 else 'stable')

PYPL Live Target Mapping (July 2026 Audit Data)

- **Vote 1 (Revenue): POSITIVE** (*Q1 2026 at +7.2% YoY acceleration vs FY2024 at +6.81%*)
- **Vote 2 (Margins): POSITIVE** (*Gross margin trend indicator marks +1*)
- **Vote 3 (FCF): NEGATIVE** (*FCF TTM down to \$5.503B vs FY2024 at \$6.768B*)
- **Vote 4 (EPS Revisions): NEGATIVE** (*Q2 guidance tracking downward to high-single digits*)
- **Vote 5 (Buybacks): POSITIVE** (*~10% of float retired TTM*)
- **Consensus Output:** 3 Positive vs 2 Negative $\rightarrow$ fd_direction = **POSITIVE**.

SECTION 4 — RECONCILED PYPL CONVICTION MATRIX

The following table contrasts the engine's current bug-ridden output against the target state once all programmatic fixes, manual calculations, and new dimensions are active:

Dimension / Component	Current Buggy State	Target State (Fixed & Fully Implemented)	Notes on State Differences
Revenue Quality	+2	+2	Correct; no changes required.
Growth Trajectory	-1	-1 <i>(or 0)</i>	Evaluate via recent quarterly YoY sequence to check weightings.
Margin Quality	0	0	Correct; R40 score (23.5) sits below the 25 threshold.
Balance Sheet	+1	+1	Correct; no changes required.
ROIC/WACC Spread	+2	+2	Correct; spread is +15.4%.
Gross Margin Trend	+1	+1	Correct; no changes required.
Debt Maturity Risk	0	0	Correct; max ceiling achieved.
CEO Quality	0	-1	New Implementation: New CEO mechanical transition penalty active.
Capital Allocation	0	+2	New Implementation: Evaluates buybacks via float reductions.

Dimension / Component	Current Buggy State	Target State (Fixed & Fully Implemented)	Notes on State Differences
Competitive Moat	0	0	New Implementation: Adversarial risk framework applied.
Macro Tailwind	0	0 (or +1)	New Implementation: Neutral consensus balance applied.
Divergence Signal	0	+2	Bug Fix: Fixed state persistence error for days_below_high.
Deal Delay Risk	0	0	Correct; consumer/SMB landscape contains no B2B pipelines.
Insider Ownership	-1	-1	Correct; tracking low institutional threshold (<1%).
Reinvestment Runway	0	+1	New Implementation: TAM/Rev ratio clears the \$10\times\$ floor.
BQ TOTAL SCORE	+4	+8 to +9	Shifting up by 4–5 core scoring points.
Entry Multiple Tax	-1	-1	Correct; EV/Rev tracks at ~1.5× (Tier 1).

Dimension / Component	Current Buggy State	Target State (Fixed & Fully Implemented)	Notes on State Differences
PE Percentile Tax	-2	0	Bug Fix: Manual calculation corrects 0× bug (PE ~10.61×).
VALUATION TAX TOTAL	-3	-1	Restores 2 points stripped by the data omission penalty.
FINAL CONVICTION	+1	+8 to +9	Shits asset out of error zone.
ENGINE VERDICT	CANCEL BUY (0%)	MAX CONVICTION (100%)	Net System Variance: +7 Conviction Points.

SECTION 5 — M&A FEATURE FLAG (Weekly Cron + SQL Storage)

Background Context

On July 15, 2026, Stripe and Advent International made an active \$60.50/share acquisition offer for PYPL. The board rejected this bid as inadequate. The engine requires a robust persistence mechanism to flag this activity, as acquisition bid premiums can distort standard divergence metrics and valuation tracking.

1. Database Layer (PostgreSQL Schema)

SQL

```
CREATE TABLE ma_activity (
    ticker    VARCHAR(20),
    bidder    VARCHAR(100),
    bid_price  DECIMAL(10,2),
    bid_date   DATE,
```

```

    board_response VARCHAR(20), -- Allowed states: 'pending' / 'rejected' / 'accepted'
    / 'lapsed'

    note          TEXT,

    last_updated  TIMESTAMP,

    active        BOOLEAN DEFAULT TRUE,

    PRIMARY KEY (ticker, bid_date)

);

```

2. Execution Pipeline (Weekly Scheduler Cron)

- **Interval:** Weekly cadence (sufficient for structural M&A changes).
- **Logic Loop:** For each ticker in the active universe, execute the following search:

Plaintext

"[company] [ticker] takeover bid acquisition offer M&A 2026"

- If evidence returns with an anchored confidence score ≥ 0.7 :
 1. Upsert records directly into the ma_activity table.
 2. Write fields to the tracking payload layer:

Python

```
conviction_store[ticker]['m_and_a_activity'] = True
```

```
conviction_store[ticker]['m_and_a_bid_price'] = bid_price
```

```
conviction_store[ticker]['m_and_a_note'] = note
```

- **Auto-Close Lifespan Boundary:** Automatically toggle active = False if the offer explicitly lapses, is finalized/accepted, or if 90 consecutive days pass with no subsequent public news updates.

3. UI Presentation Tier

While active = True, parse the string payload and display a prominent alert on the main asset conviction dashboard view:

 **ACTIVE M&A:** \$60.50/share offer from Stripe/Advent (board: inadequate)

SECTION 6 — USER INTERFACE ACTIONS (Forward to Parth)

Please route these three action-handler bugs directly to Parth for immediate refactoring. The interface event triggers are declared but disconnected from their navigation routers:

1. **Drawer Header Link:** The "Full FS view →" link inside the drawer component currently performs no action. Wire its click event handler to route users to the comprehensive deep-dive Financial Strength page (/fs-depth).
2. **Signal Row BQ Action:** The "BQ drill" interactive text link in the core signal matrix row fails to fire. Wire its click handler to open the tracking drawer defaulted to the **BQ Tab**.
3. **Signal Row FS Action:** The "FS page →" action layout node in the signal row fails to fire. Wire its click handler to route navigation to the full parent Financial Strength page (/fs).

Storage Architecture Note for Parth's UI: Ensure all **15 individual BQ sub-components** are systematically stored as standalone key-value entries inside conviction_store per ticker (instead of solely saving the aggregate BQ_RAW score). The frontend drawer component requires individual sub-component tracking keys to correctly render the interactive UI breakdown drilldowns.

IMPLEMENTATION PRIORITY PATHWAY

P1: DATA & SYSTEM BUG FIXES	
1. PE manual calculation cascade (elim. -2 default)	
2. FCF OEY calculation cascade	
3. Persist and log divergence 'days_below_high' counter	

| P2: MISSING BQ DIMENSIONS |

| 4. Capital Allocation (Dim 9 via float tracking) |

| 5. Programmatic 5 Fundamental Direction (FD) Votes |

| 6. CEO Quality (Dim 8 quantitative + mechanical rule) |

| 7. Competitive Moat (Dim 10 adversarial prompt matrix) |

| 8. Macro Tailwind & Reinvestment Runway (Dims 11 & 15) |



| P3 & P4: CORE ENGINE FEATURES & UI |

| 9. Database M&A schema implementation & Weekly Cron |

| 10. Connect Parth's UI navigation click handlers |
